

Unitary and Orthosymplectic 3d $\mathcal{N} = 4$ Quiver Gauge Theories

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Overview

- ① Supersymmetry
- ② Quivers
- ③ Moduli spaces
 - Hilbert series
 - Monopole formula
 - Molien-Weyl formula
- ④ $U(1)$ gauge theory with N flavours
 - Coulomb branch
 - Higgs branch
- ⑤ $Sp(1) \times Sp(2)$ gauge theory with 4 flavours
 - Coulomb branch
 - Higgs branch
- ⑥ Conclusion

Supersymmetry

- Symmetries are central object of study in theoretical physics
- Can introduce *supersymmetries* to link bosonic and fermionic fields (allows SQCD, superstring theory and supergravity)

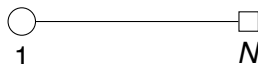
$$Q|\text{fermion}\rangle = |\text{boson}\rangle, \quad Q|\text{boson}\rangle = |\text{fermion}\rangle$$

- Fields now packaged into supermultiplets [1] (in our case vector multiplets and hypermultiplets)

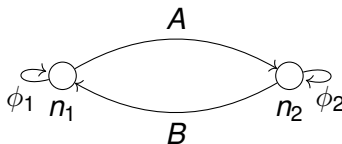
$$V(\phi) = \sum_i \left| \frac{\partial W}{\partial \phi_i} \right|^2$$

Quivers

- Supersymmetric actions can get very messy and complicated
- Difficult to determine important details such as symmetries and dynamics
- Use quivers to encode symmetry information of theory [2]



- Can easily read off *superpotential* [3]



$$W = \text{tr}(AB\phi_1 - BA\phi_2)$$

- Due to scalars vacuum is non-trivial [4]
- Define *moduli space* of inequivalent vacua
- Vacuum expectation values parameterise moduli space [5]
- Moduli space of quiver gauge theories naturally decomposes into two branches [6]
- *Coulomb branch* from vector multiplet contribution
- *Higgs branch* from hypermultiplet contribution
- Quaternionic dimension easy to read off from quivers [7]

Moduli spaces

Hilbert series

- Moduli space is naturally a variety
- Useful tool to characterise geometric spaces
- Can extract information about moduli spaces from Hilbert series [8]
- Let $\mathcal{V}$ be an algebraic variety over the graded polynomial ring R

$$HS(t) = \sum_i \dim(R_i) t^i$$

- Complex dimension can be read off from Hilbert series [9]

$$HS(t) = \frac{Q(t)}{(1-t)^d}$$

Moduli spaces

Monopole formula

- Use monopole formula to count dressed monopole operators at each conformal dimension [7, 10]

$$\begin{aligned} \text{HS}(t) &= \sum_{m \in \Gamma/\mathcal{W}} t^{2\Delta(m)} P(t, m) \\ &= \sum_{\vec{m}_1} \sum_{\vec{m}_2} \cdots \sum_{\vec{m}_x} t^{2\Delta(\vec{m}_1, \vec{m}_2, \dots, \vec{m}_x)} \prod_{i=1}^x P_{G_i}(t, \vec{m}_i) \end{aligned}$$

- 1 Determine classical dressing factors
- 2 Compute conformal dimension $\Delta = \Delta_{\text{vec}} + \Delta_{\text{hyp}}$ [11]
- 3 Perform sums

Moduli spaces

Molien-Weyl formula

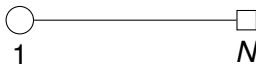
- Use Molien-Weyl formula to determine the F-flat Hilbert series the Higgs branch and then quotient by the gauge group action [4, 7]

$$HS_{\mathcal{H}}(t) = \int_G d\mu_G \frac{PE \left[\sum_{i=1}^{2N_{hyp}} \chi_{R_i}^{G_i}(\omega) \chi_{R'_i}^{G'_i}(\omega') t \right]}{PE \left[\sum_{j=1}^{N_{vec}} \chi_{R''_j}^{G_j}(\omega'') t^2 \right]}$$

- 1 Compute characters for the representations of vector multiplets and hypermultiplets
- 2 Determine Haar measures
- 3 Perform integrals

$U(1)$ gauge theory with N flavours

- Consider SQED



- Can see $\dim_{\mathbb{H}} \mathcal{C} = 1$ and $\dim_{\mathbb{H}} \mathcal{H} = N - 1$

$U(1)$ gauge theory with N flavours

Coulomb branch

- Determine Hilbert series using monopole formula

$$\text{HS}_{\mathcal{C}} = \frac{1 + t^N}{(1 - t^2)(1 - t^N)}$$

- $\dim_{\mathbb{C}} \mathcal{C} = 2 = 2 \dim_{\mathbb{H}} \mathcal{C}$ as expected
- Can match number of operators at each order to find Coulomb branch
- For $N = 2$ get orbifold $\mathbb{C}^2 / \mathbb{Z}_2$
- Global symmetry is $U(1)$ except when $N = 2$ then $SU(2)$

$U(1)$ gauge theory with N flavours

Higgs branch

- Determine Hilbert series using Molien-Weyl formula

$$HS_{\mathcal{H}} = \frac{\sum_{k=0}^{N-1} \binom{N-1}{k}^2 t^{2k}}{(1-t^2)^{2(N-1)}}$$

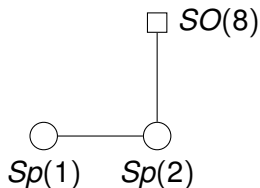
- $\dim_{\mathbb{C}} \mathcal{H} = 2(N-1) = 2\dim_{\mathbb{H}} \mathcal{H}$ as expected
- Hilbert series too complicated to extract Higgs branch directly
- Use superpotential $W = \text{tr}(BA\phi)$ with condition $\frac{\partial W}{\partial \phi} = BA = 0$
- Now defining $M = AB$ gives condition $M^2 = 0$ and

$$\mathcal{H} = \{M_{N \times N} | \text{tr} M = 0, M^2 = 0, \text{rank } M \leq 1\}$$

- Global symmetry is $SU(N)$

$Sp(1) \times Sp(2)$ gauge theory with 4 flavours

- Now consider more exotic orthosymplectic theory

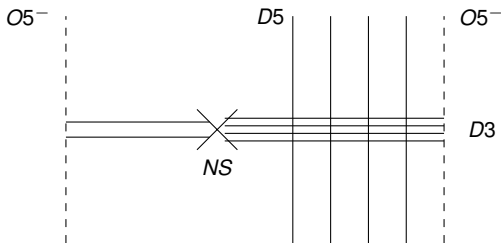


- Can see $\dim_{\mathbb{H}} \mathcal{C} = 3$ and $\dim_{\mathbb{H}} \mathcal{H} = 11$

$Sp(1) \times Sp(2)$ gauge theory with 4 flavours

Coulomb branch

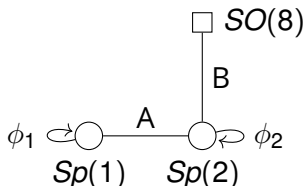
- Try to compute Hilbert series using monopole formula but diverges
- Compute Higgs branch of magnetic quiver using Molien-Weyl formula
- Realise Sp gauge symmetry using orientifold planes [12]



$Sp(1) \times Sp(2)$ gauge theory with 4 flavours

Higgs branch

- Global symmetry is $Sp(1) \times SO(8)$
- Use Molien-Weyl formula again to determine Hilbert series
- For orthosymplectic quivers edges only represent half-hypermultiplets [13]



- Use Mathematica due to difficulty of integration

Conclusion

- Computed properties of SQED $U(1)$ quiver
- Analysed related $U(2)$ gauge theory
- Extracted information from $Sp(1) \times Sp(2)$ gauge theory
- Planning to find magnetic quiver for $Sp(1) \times Sp(2)$ gauge theory to compute Coulomb branch

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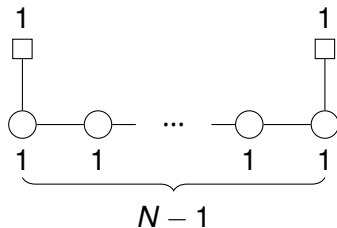
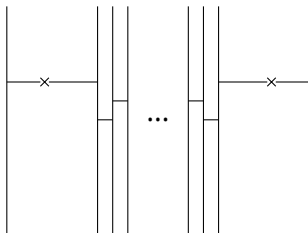
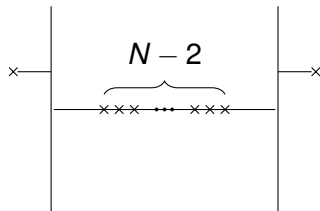
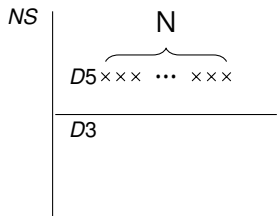


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Brane manipulations



$U(1)$ gauge theory with N flavours

Related theories

- Consider similar theory where Higgs branch rank $M \leq 2$



- Also have mirror theory where Coulomb branch has rank $M \leq 2$

